# Results: Language, Topics, and Embeddings

## RQ3: Topical and Linguistic Structure I

Text analysis operates over titles, abstracts, and (when available) full text. A TF-IDF representation over a 137-feature vocabulary feeds non-negative matrix factorization, which extracts 5 latent topics cross-cutting the subfield taxonomy. The top vocabulary terms are: atmospheric, uncertainty, spectra, temperature, abundance, analysis, assumptions, opacity, retrievals, sources, stellar, coverage, evaluated, observed, across, composition, models, provide, competing, retrieval.

**Table 3. NMF topics extracted from the corpus.**

# RQ3: Topical and Linguistic Structure I

| Topic | Top terms |
|---|---|
| 0 | atmospheric, cloud, retrieved, species, changes, structure, models, explore |
| 1 | prior, methane, depend, spectral, dioxide, simulated, carbon, constrained |
| 2 | contributions, separate, resolution, planetary, high, systematics, modelled, instrument |
| 3 | population, reporting, studies, preprocessing, require, harmonized, separates, specific |
| 4 | model, intervals, misspecification, noise, quantify, effect, record, dependencies |

## RQ3: Topical and Linguistic Structure I

The topic labels are computed from the retained corpus and are not hand-assigned. Their interpretation should be checked against the generated topic-term weights and the configured subfield taxonomy; fixture-derived topics describe synthetic text patterns, not the scientific field.

# RQ3: Topical and Linguistic Structure I

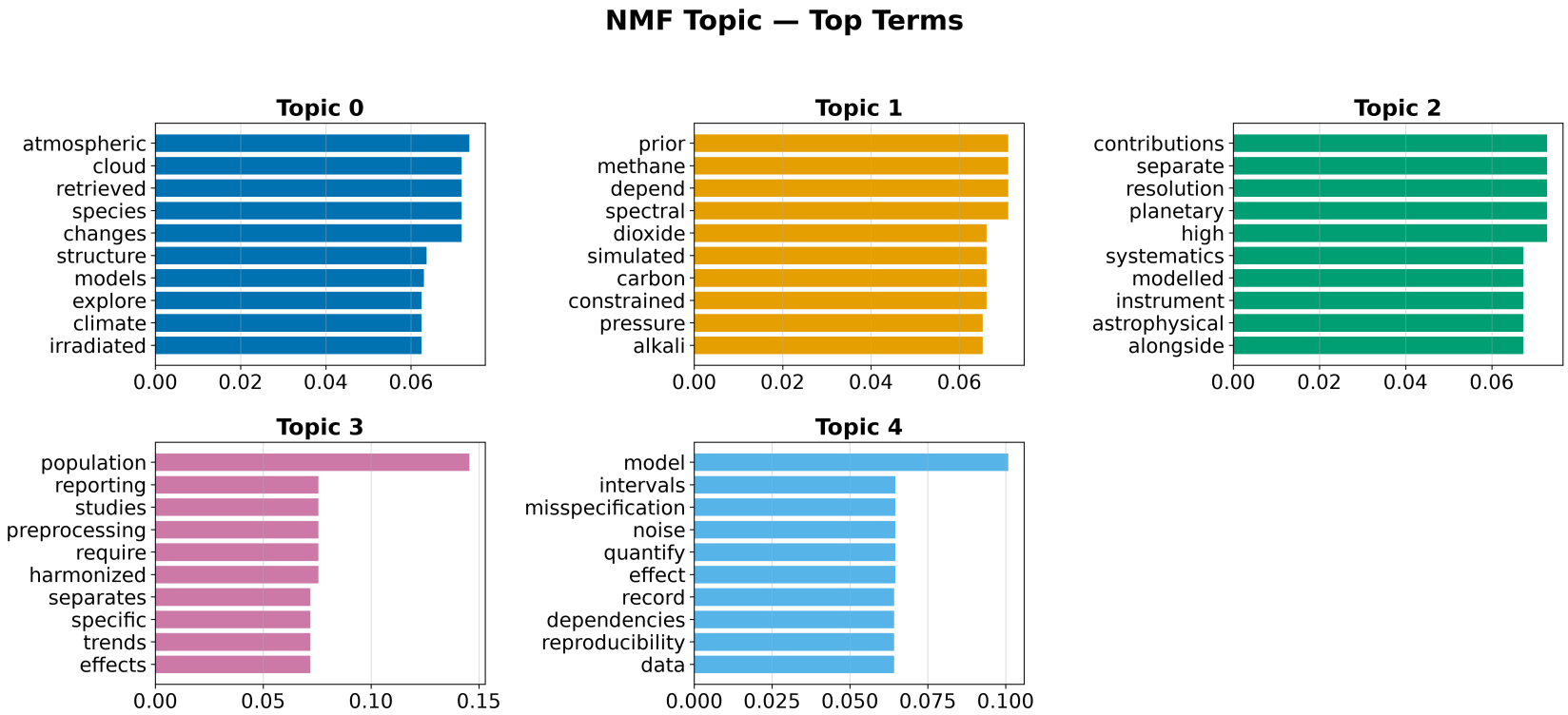


Figure 8: Topic-term bar charts for Exoplanet Atmospheres. Each panel shows the top weighted terms for one of 5 NMF topics, with bar length proportional to the topic-term weight in the $\mathbf{H}$ matrix.

## Document Embeddings I

Offline deterministic embeddings (TF-IDF followed by truncated SVD) place every document in a shared 50-dimensional vector space. Embedding the same text twice yields identical vectors, so the derived similarity matrix, nearest-neighbour lists, clusters, and two-dimensional projection are all reproducible.

# Document Embeddings I

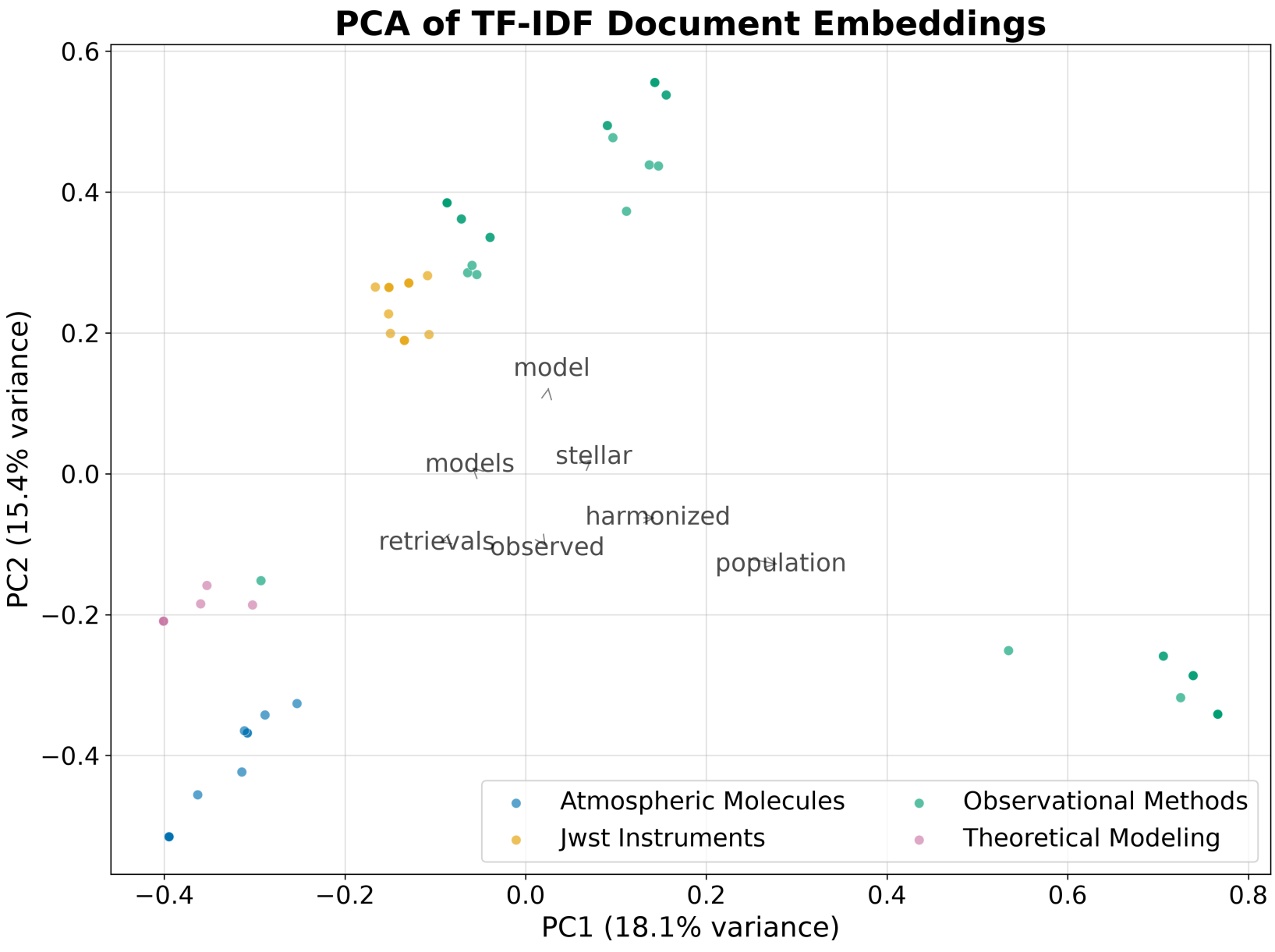


Figure 9: PCA projection of document embeddings for Exoplanet Atmospheres. Each point represents one document projected onto the first two principal components of the TF-IDF/SVD embedding. Colours indicate subfield assignment, showing how the topical geography relates to the keyword taxonomy.

# Document Embeddings I

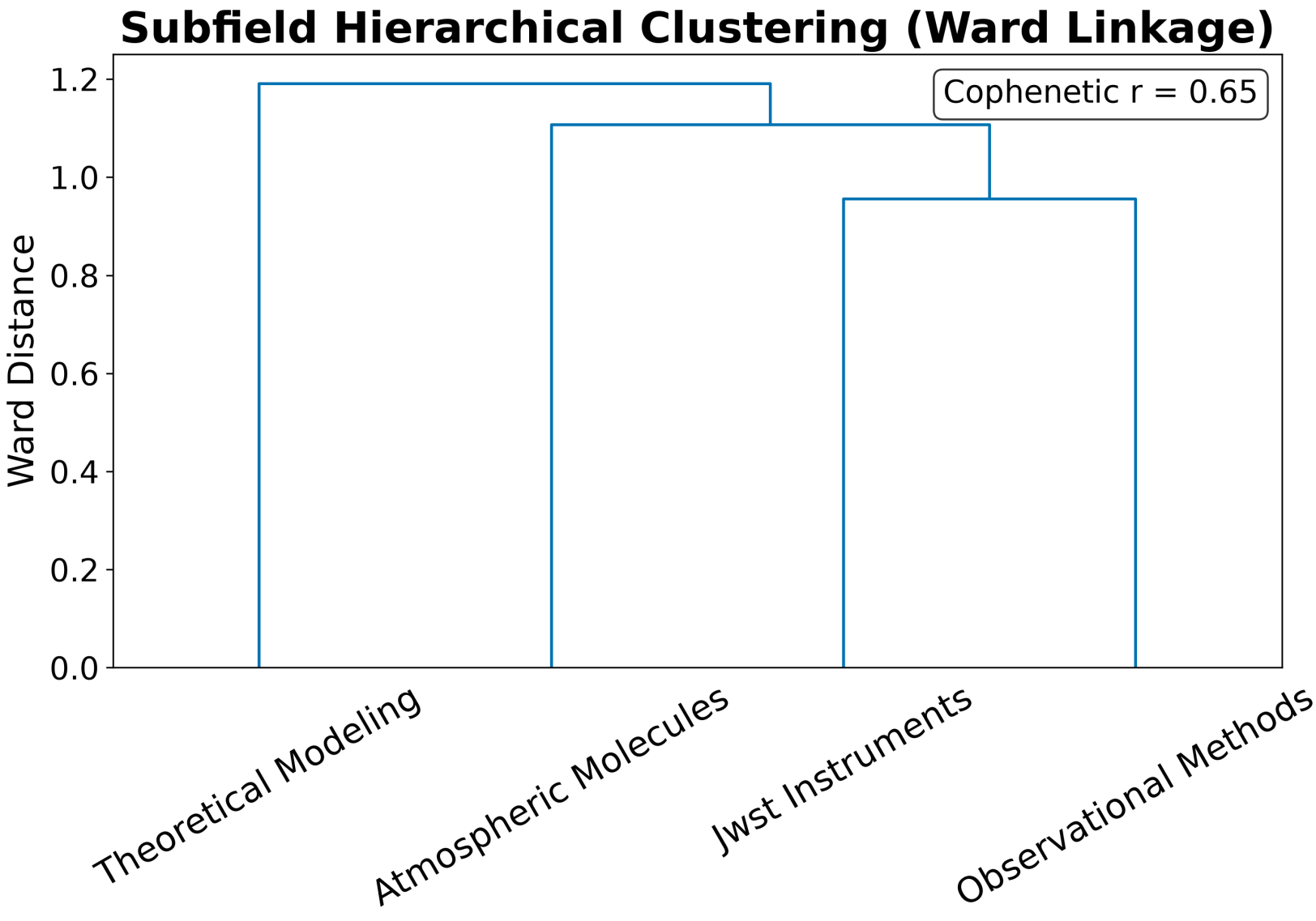


Figure 10: Hierarchical clustering dendrogram of document embeddings. The tree shows the similarity structure of the corpus: documents that join low in the tree are semantically similar, while high-level splits separate the major topical clusters.

# Term Analysis I

The TF-IDF term heatmap reveals which terms discriminate between subfields: terms with high between-subfield variance (rather than high global mean) are selected for display.

# Term Analysis I

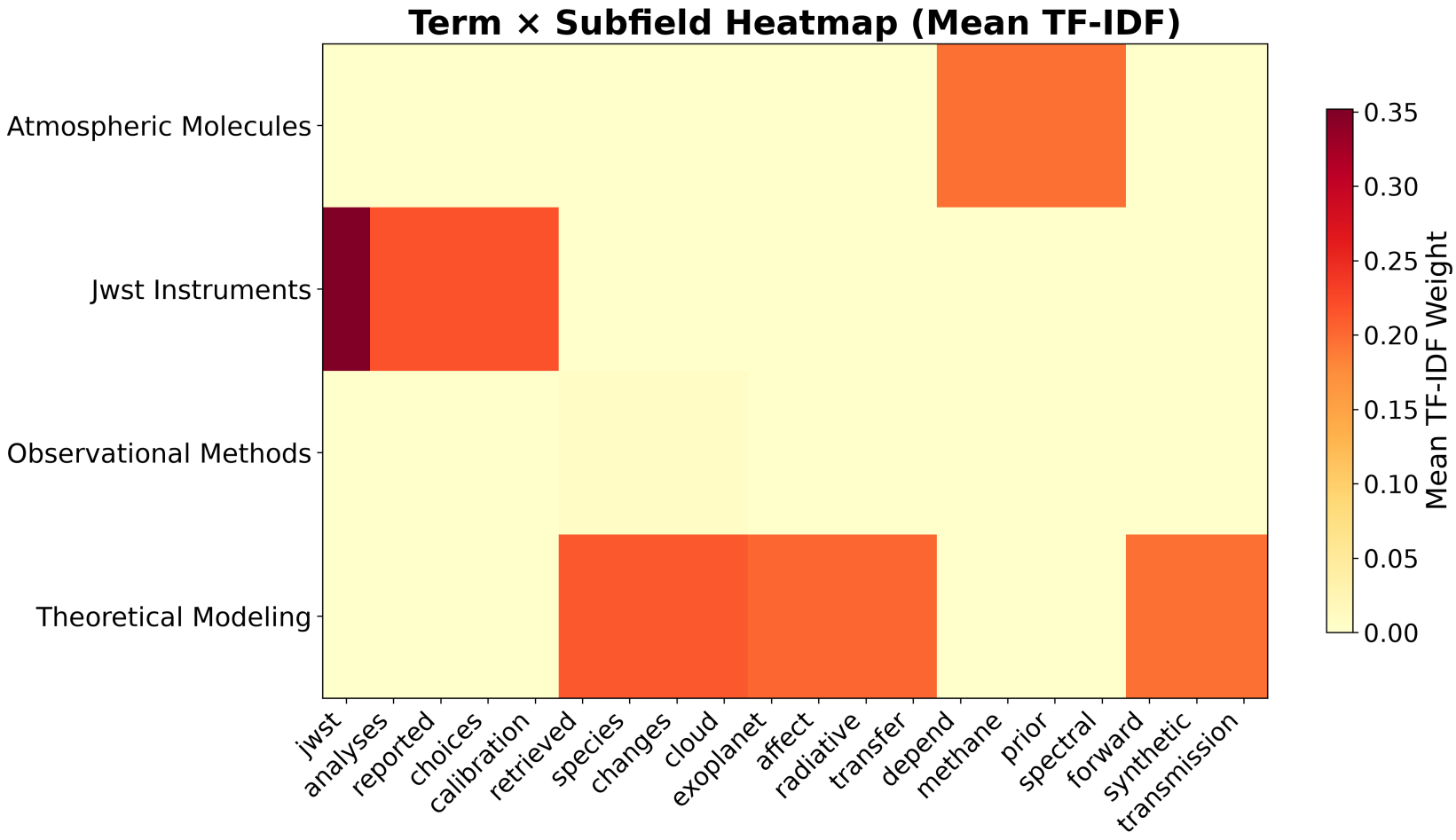


Figure 11: Term heatmap for Exoplanet Atmospheres. Each cell shows the mean TF-IDF weight of a term within a subfield. Terms are selected by between-subfield variance to highlight discriminative vocabulary rather than globally frequent terms.

# Term Analysis I

Named entity extraction over the 80 abstracts identified 1 unique entities. The most frequent entities reflect the retained corpus and the configured extraction rules; source coverage and fixture status determine how they may be interpreted.

**Table 4. Top named entities in abstracts.**

# Named Entity Analysis I

| Entity | Frequency |
|---|---|
| JWST | 17 |

# Named Entity Analysis I

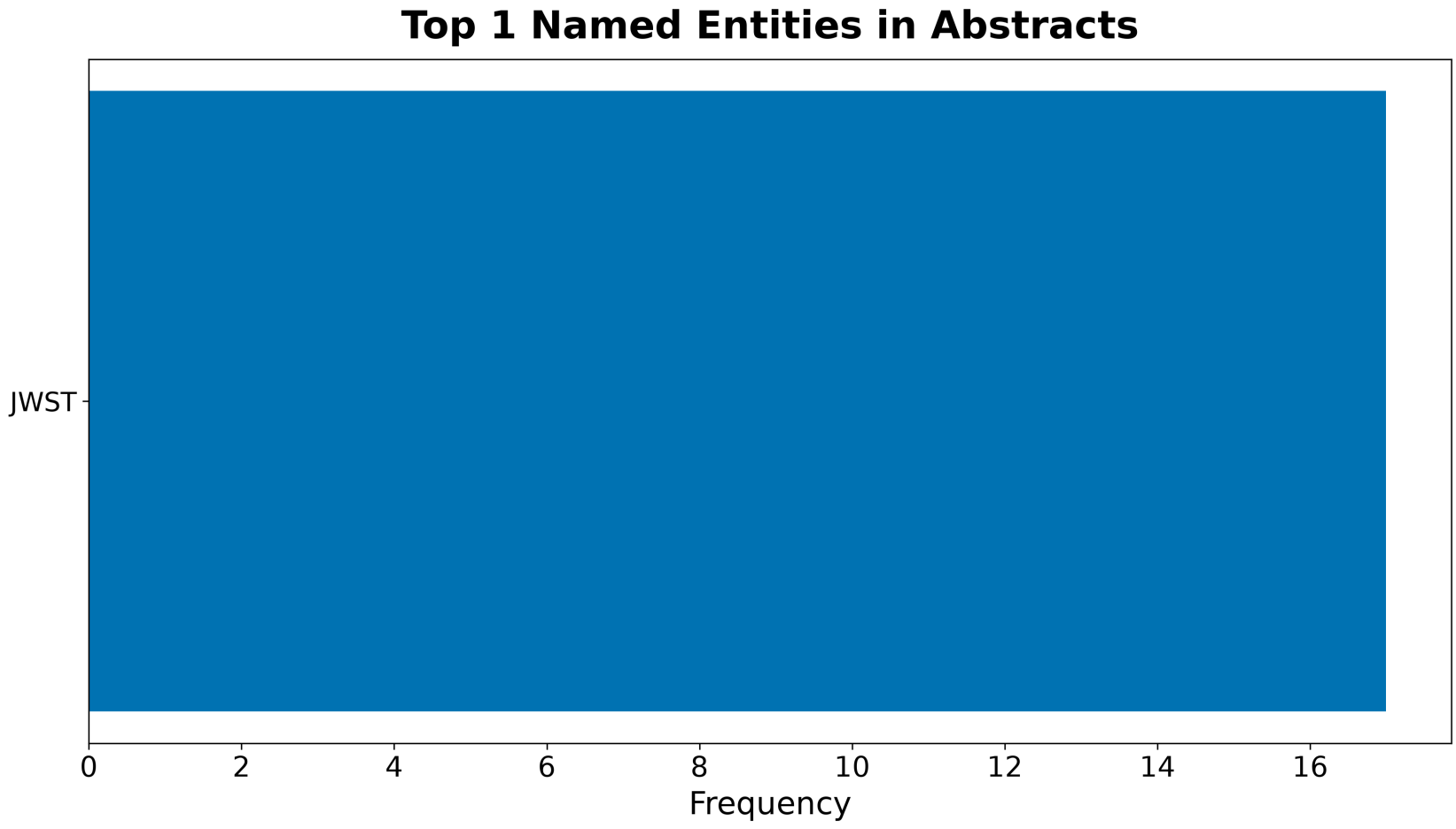


Figure 12: Top named entities for Exoplanet Atmospheres. The horizontal bar chart shows the most frequently extracted entities from abstracts, revealing recurring objects, instruments, methods, and concepts in the retained corpus.

# Named Entity Analysis I

# Named Entity Analysis II

**Table 5. Top keyphrases by TF-IDF score.**

# Named Entity Analysis I

| Keyphrase | Score |
|---|---|
| jwst | 0.0714 |
| population | 0.0556 |
| analyses | 0.0435 |
| calibration | 0.0435 |
| calibration choices | 0.0435 |
| choices | 0.0435 |
| combined | 0.0435 |
| jwst analyses | 0.0435 |

# Named Entity Analysis I

| Keyphrase | Score |
|---|---|
| models | 0.0435 |
| models calibration | 0.0435 |
| models calibration choices | 0.0435 |
| model | 0.0400 |
| atmospheric | 0.0392 |
| abundance | 0.0385 |
| assumptions | 0.0385 |

The TF-IDF/SVD embeddings place every document in a 50-dimensional vector space. K-means clustering with $k = 5$ clusters partitions the corpus into topically coherent groups. The top similar document pairs, ranked by cosine similarity, reveal the most closely related works in the corpus.

**Table 6. Top 10 most similar document pairs.**

# Embedding Similarity and Clustering I

| Paper A | Paper B | Similarity |
|---|---|---|
| doi:10.5555/exoplanet atmospheres.003 1 | doi:10.5555/exoplanet atmospheres.005 5 | 1.0000 |
| doi:10.5555/exoplanet atmospheres.004 6 | doi:10.5555/exoplanet atmospheres.007 0 | 1.0000 |
| doi:10.5555/exoplanet atmospheres.005 5 | doi:10.5555/exoplanet atmospheres.003 1 | 1.0000 |
| doi:10.5555/exoplanet atmospheres.007 0 | doi:10.5555/exoplanet atmospheres.004 6 | 1.0000 |
| doi:10.5555/exoplanet atmospheres.000 0 | doi:10.5555/exoplanet atmospheres.002 4 | 1.0000 |
| doi:10.5555/exoplanet atmospheres.000 0 | doi:10.5555/exoplanet atmospheres.004 8 | 1.0000 |
| doi:10.5555/exoplanet atmospheres.000 0 | doi:10.5555/exoplanet atmospheres.007 2 | 1.0000 |
| doi:10.5555/exoplanet atmospheres.000 2 | doi:10.5555/exoplanet atmospheres.004 4 | 1.0000 |

# Embedding Similarity and Clustering I

| Paper A | Paper B | Similarity |
|---|---|---|
| doi:10.5555/exoplanet atmospheres.0002 | doi:10.5555/exoplanet atmospheres.0020 | 1.0000 |
| doi:10.5555/exoplanet atmospheres.0002 | doi:10.5555/exoplanet atmospheres.0068 | 1.0000 |

# Embedding Similarity and Clustering I

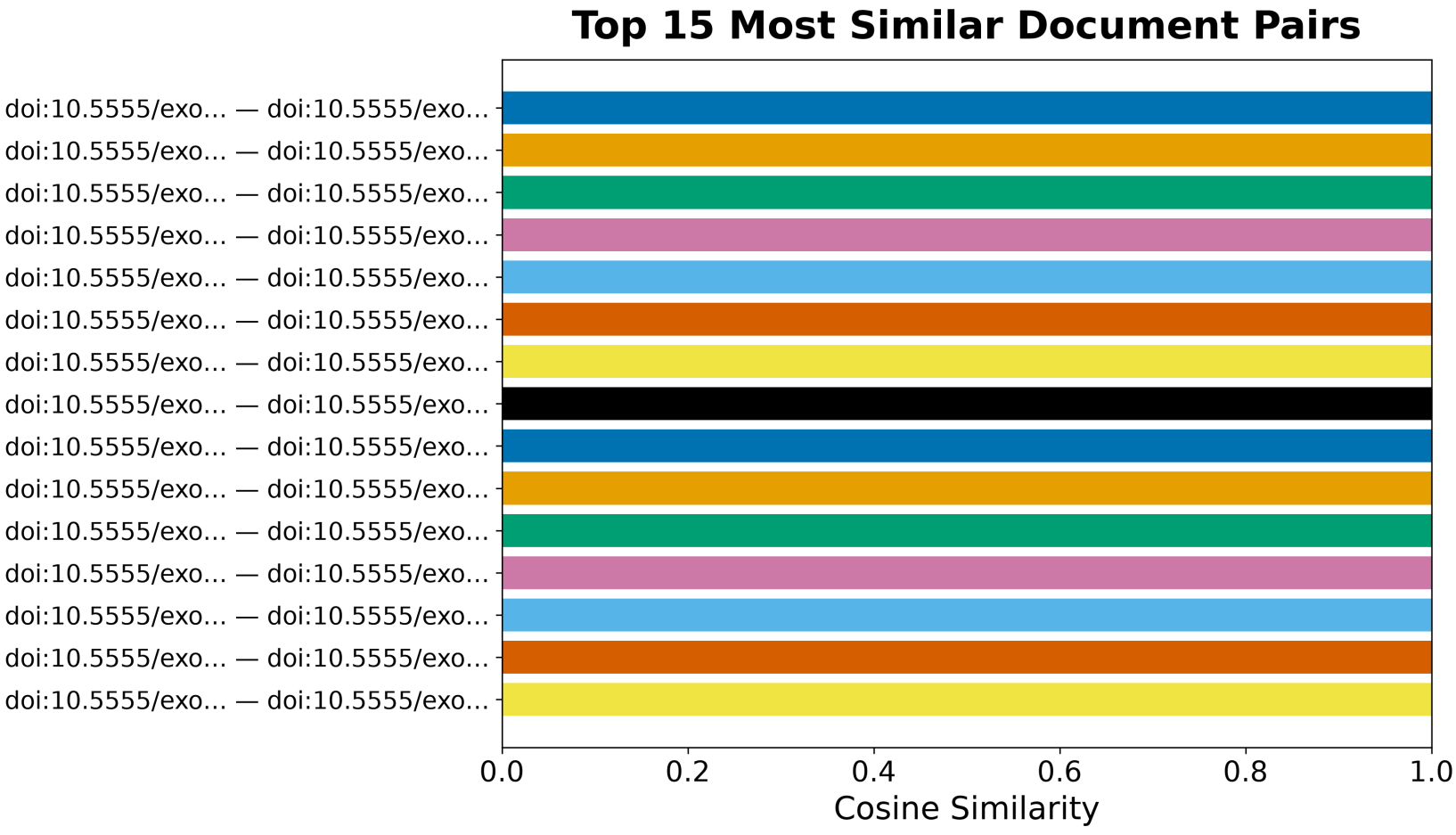


Figure 13: Document similarity for Exoplanet Atmospheres. The horizontal bar chart shows the 15 most similar document pairs ranked by cosine similarity of their TF-IDF/SVD embeddings. High-similarity pairs share topical and lexical content.

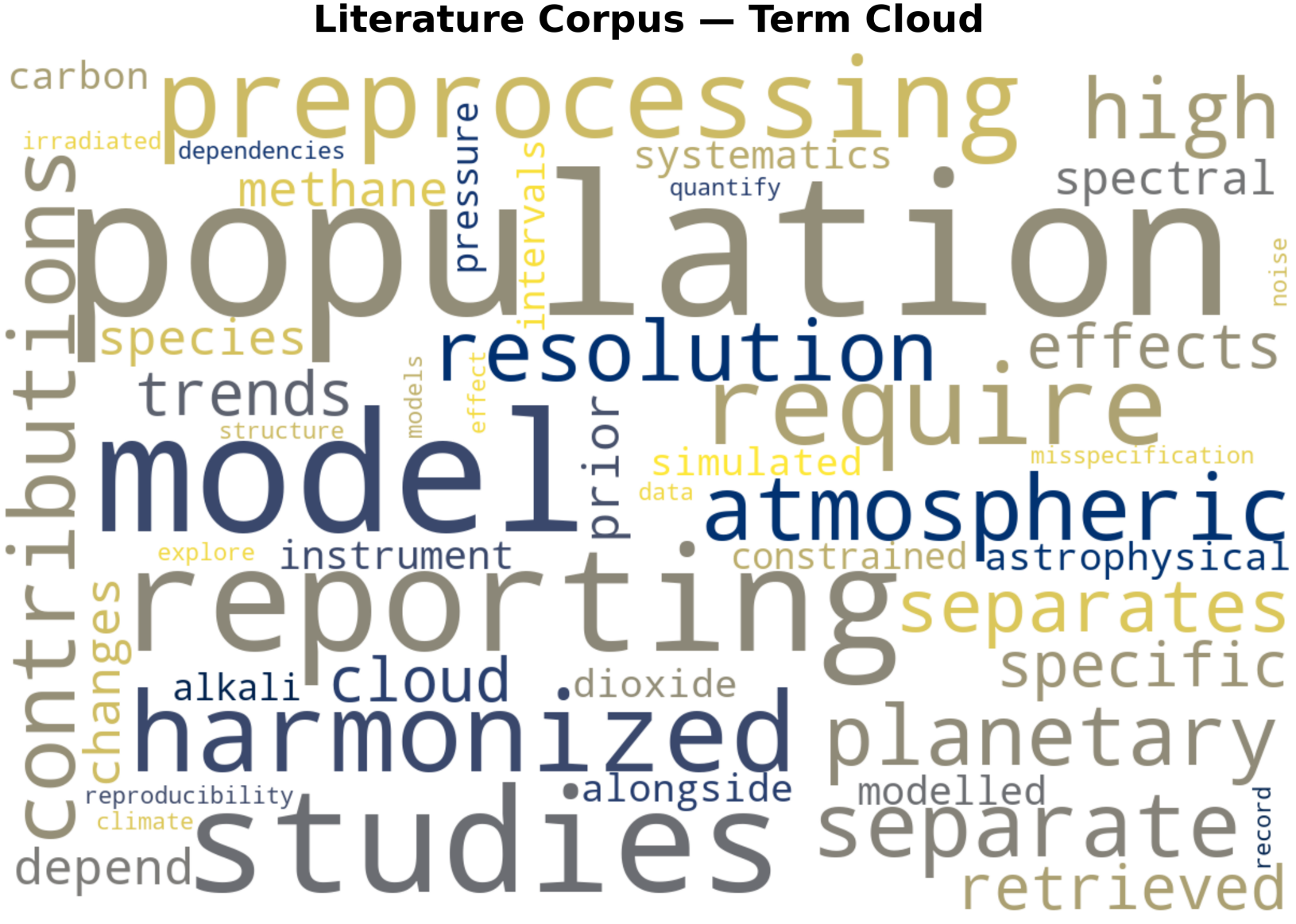

Literature Corpus — Term Cloud
carbon
preprocessing
high
irradiated
dependencies
systematics
spectral
methane
quantify
pressure
intervals
population
noise
species
resolution
effects
trends
require
structure
models
effect
prior
simulated
misspecification
data
model
atmospheric
explore
instrument
constrained
astrophysical
reporting
separates
changes
specific
alkali
cloud
dioxide
harmonized
planetary
reproducibility
alongside
modelled
climate
separate
record
depend
studies
retrieved
contributions

Figure 14: Term cloud for Exoplanet Atmospheres. Term sizes are proportional to their TF-IDF weights across the corpus, providing a visual summary of the dominant vocabulary.

# Embedding Similarity and Clustering I

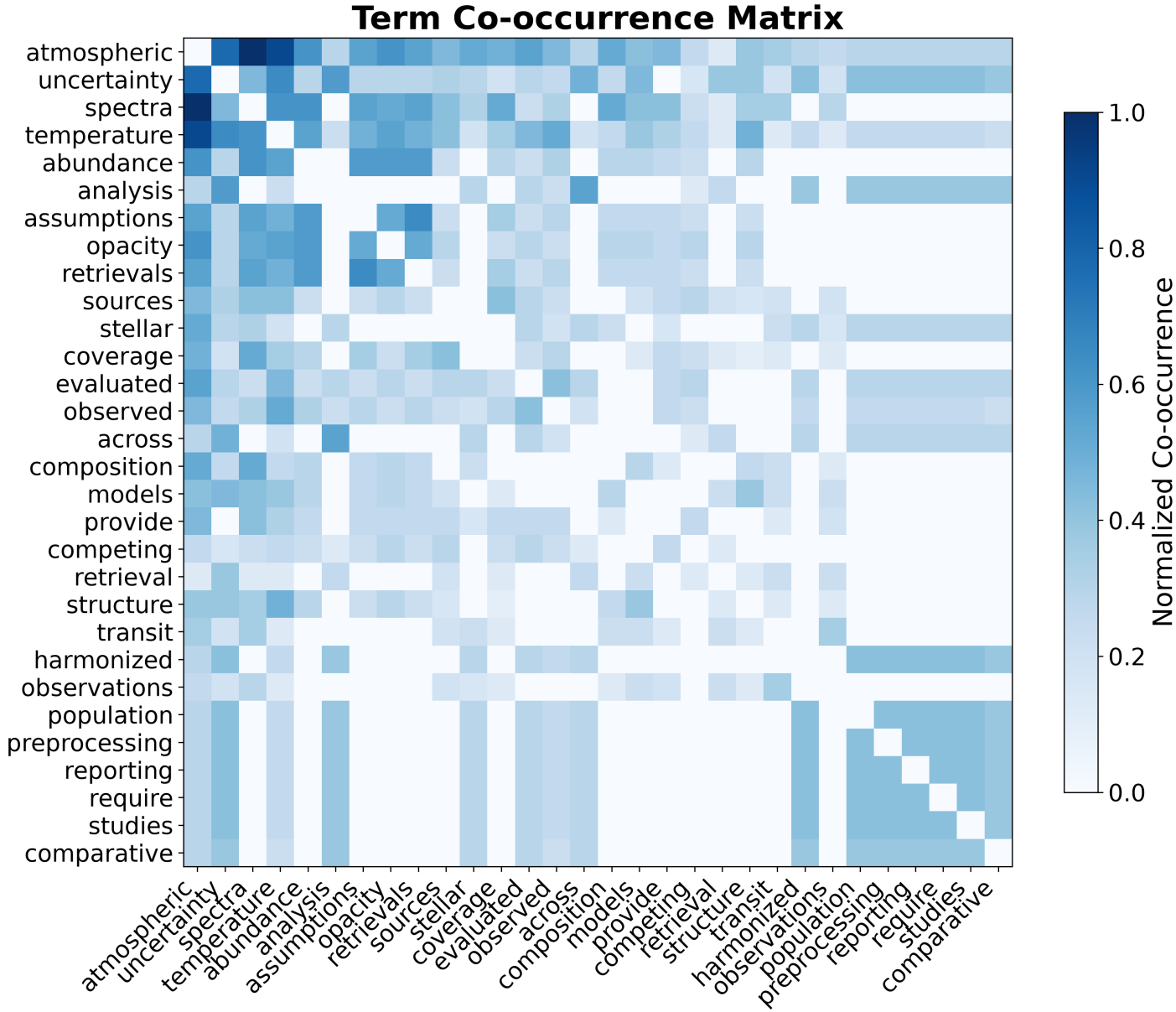


Figure 15: Term co-occurrence matrix for Exoplanet Atmospheres. Each cell shows the normalized co-occurrence frequency of two terms within the same document, revealing which concepts tend to appear together in the literature.

## Embedding Similarity and Clustering I

These embeddings support semantic retrieval over the corpus and the visual map of the literature's topical geography.